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(54) STAIR-CLIMBING MACHINE AND RESISTANCE CONTROLLING DEVICE THEREOF

(71) Applicant: REXON INDUSTRIAL CORP., LTD., Taichung (TW)

(72) Inventor: Wen-Tu LI, Taichung (TW)

(73) Assignee: REXON INDUSTRIAL CORP., LTD., Taichung (TW)

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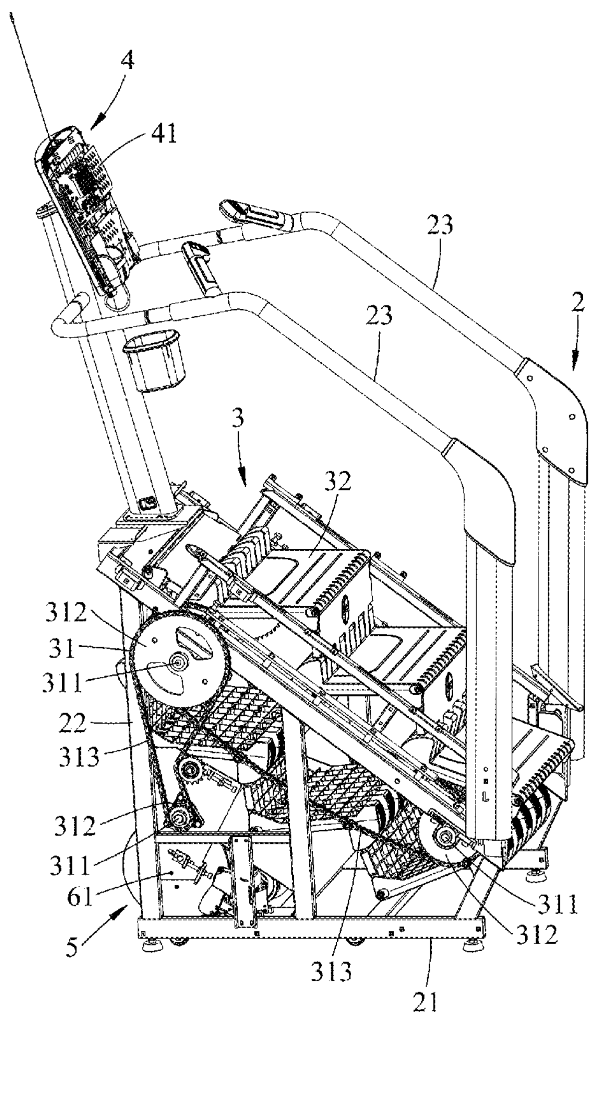
(57) ABSTRACT

A resistance controlling device includes a flywheel and a damper unit. The damper unit includes a damper rack and a plurality of permanent magnets disposed on the damper rack. The damper rack is movable between a high-resistance position, where the damper rack is proximate to the flywheel, and a low-resistance position, where the damper rack is distal from the flywheel. When the flywheel is driven to rotate as the damper rack is in the high-resistance position, the permanent magnets exert a first resistance force on the flywheel. When the flywheel is driven to rotate as the damper rack is in the low-resistance position, the permanent magnets exert a second resistance force weaker than the first resistance force on the flywheel. A stair-climbing machine includes a base frame unit, a transmission device, a control device and the abovementioned resistance controlling device.



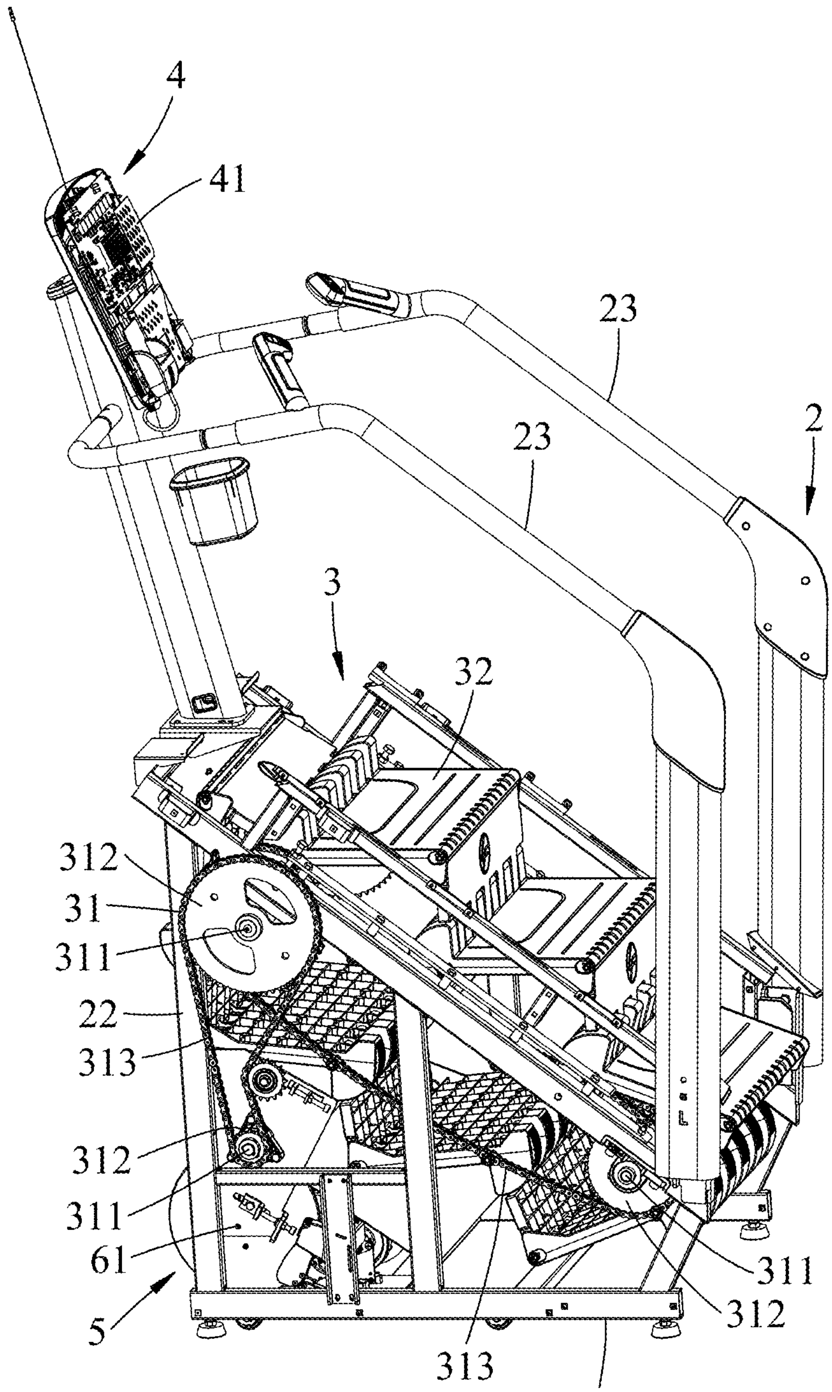


FIG. 1

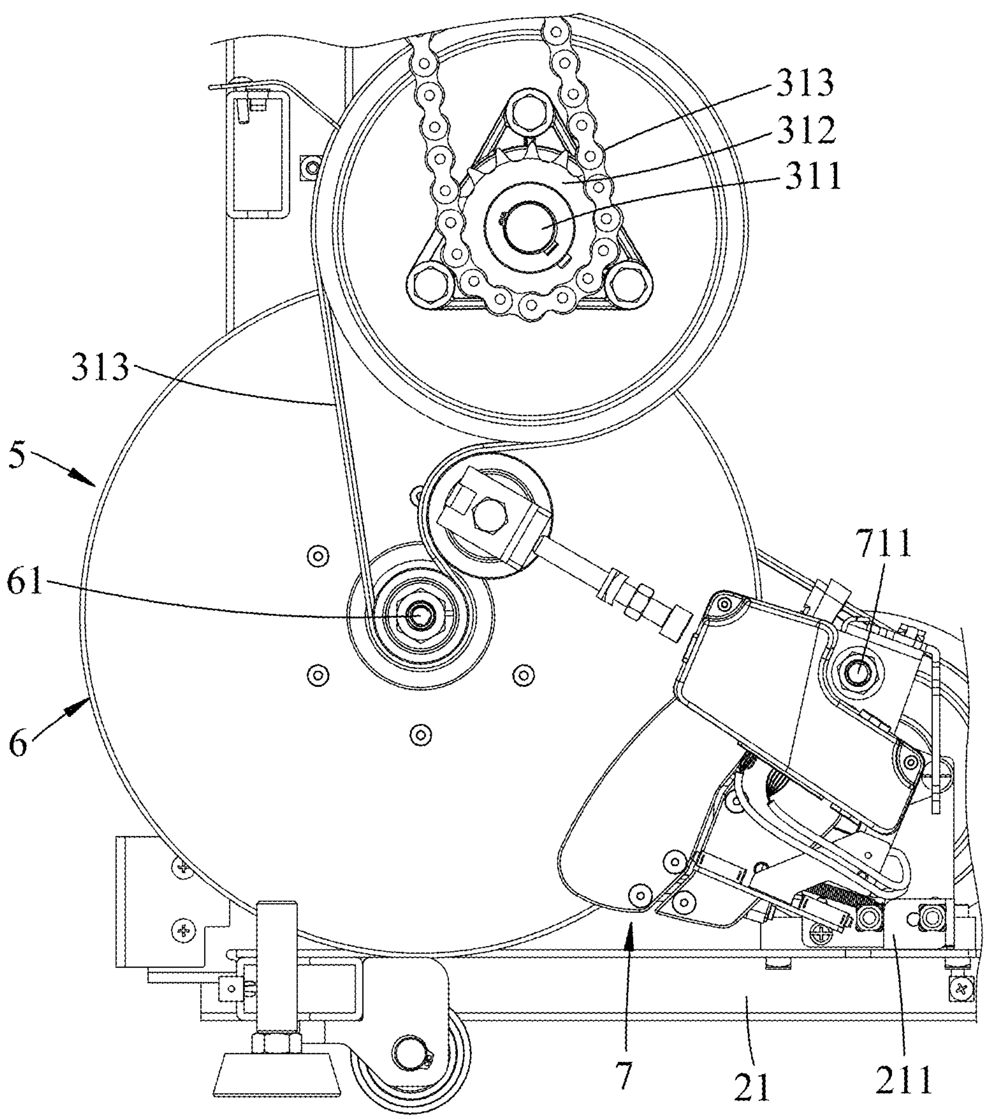


FIG. 2

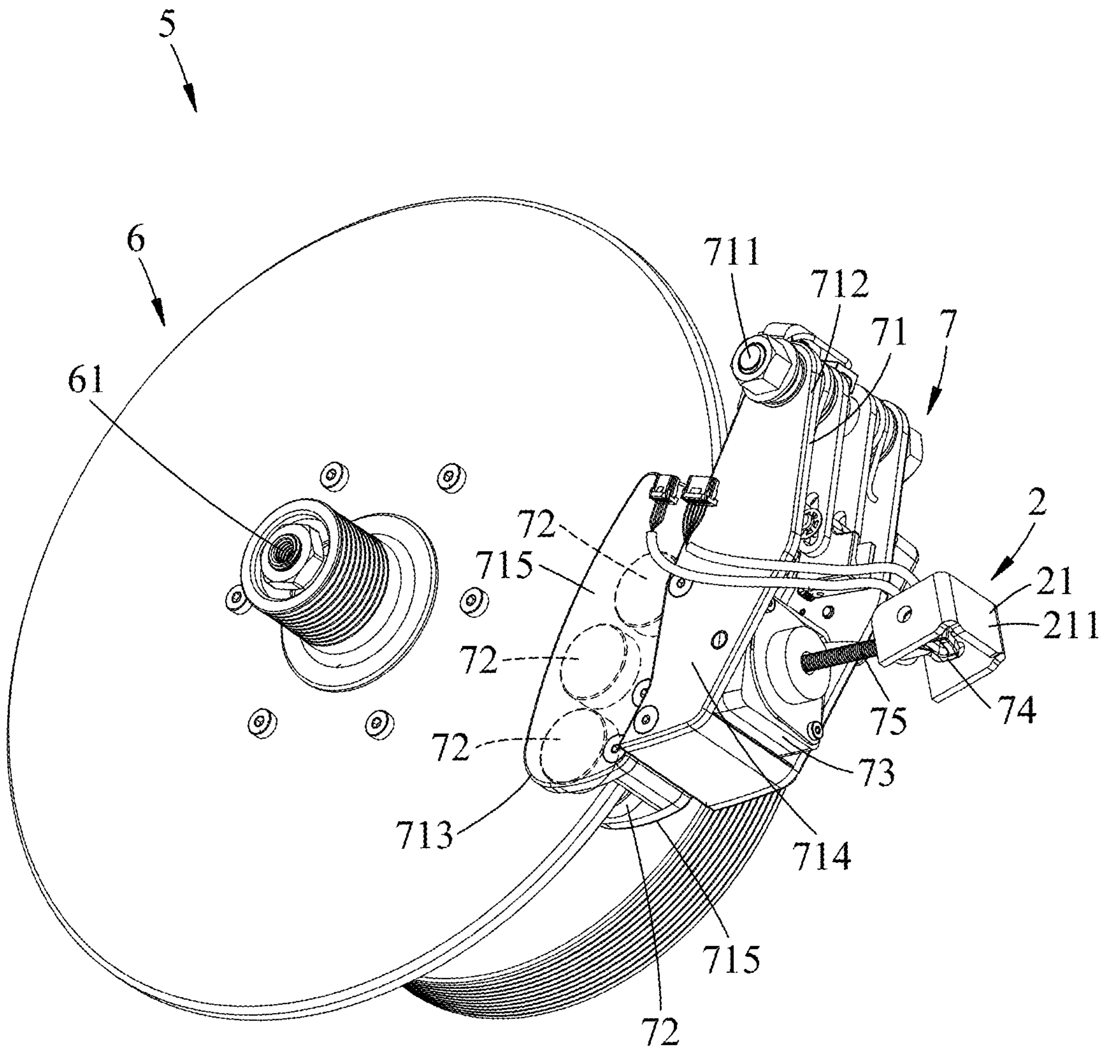


FIG. 3

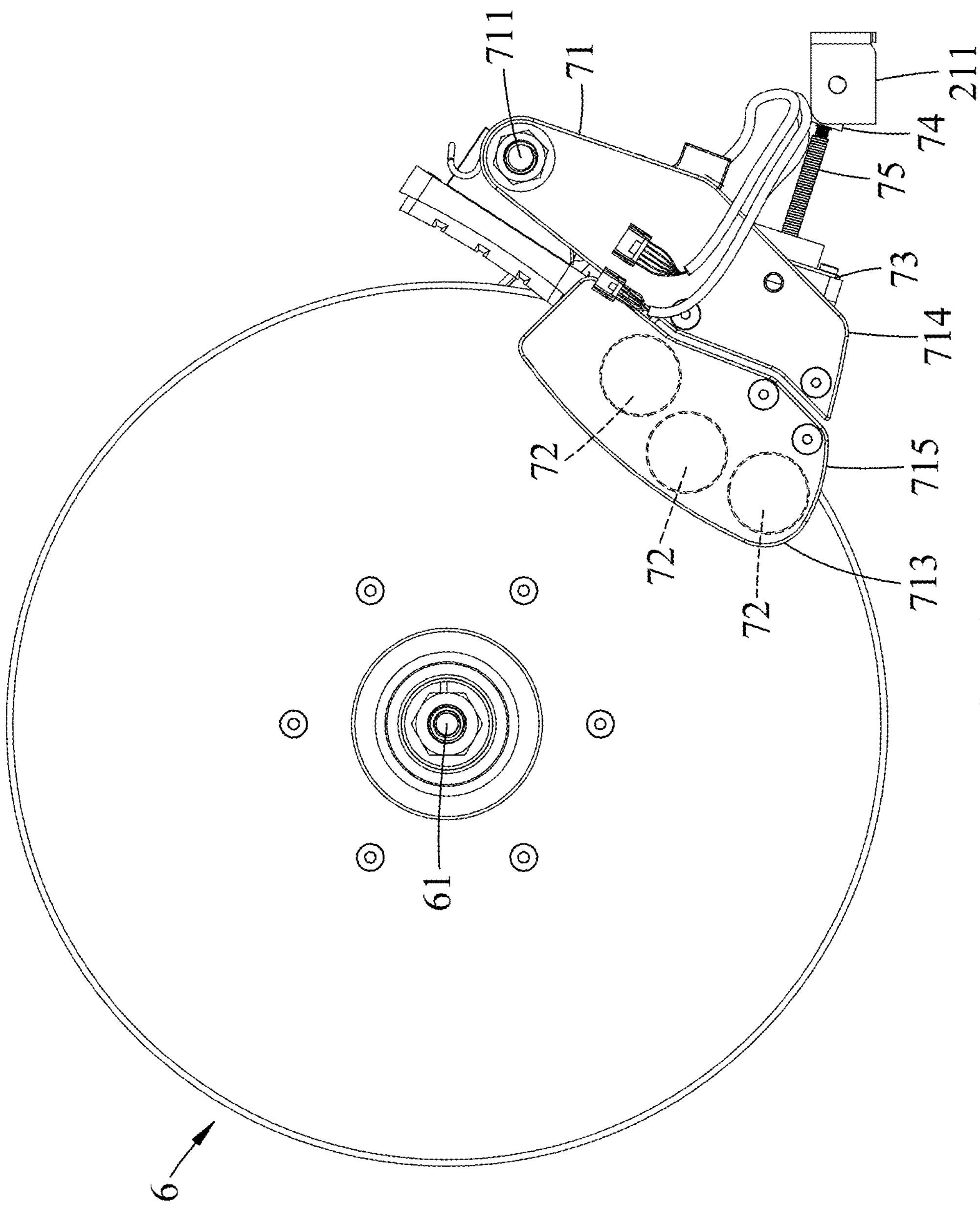


FIG. 4

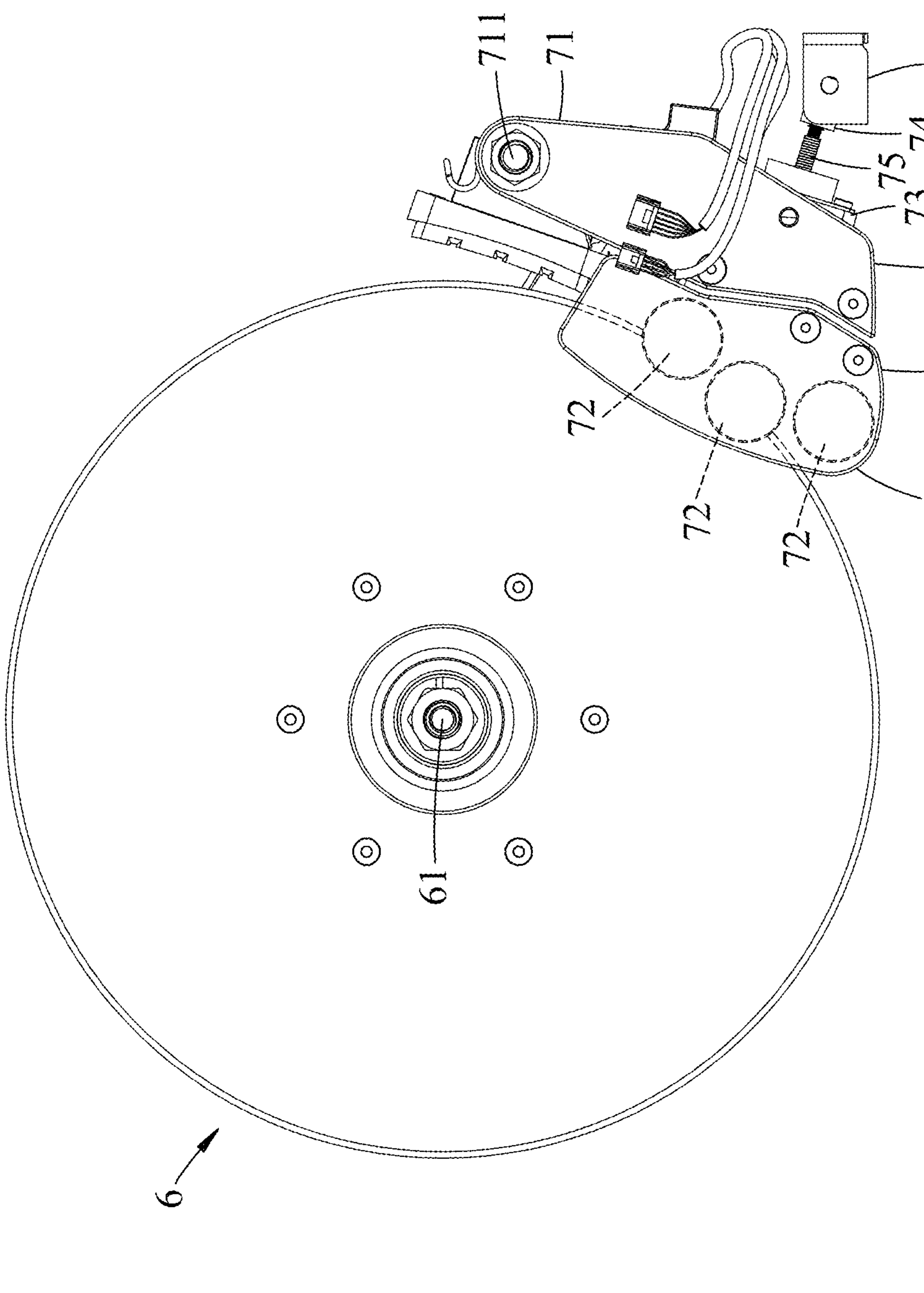


FIG. 5

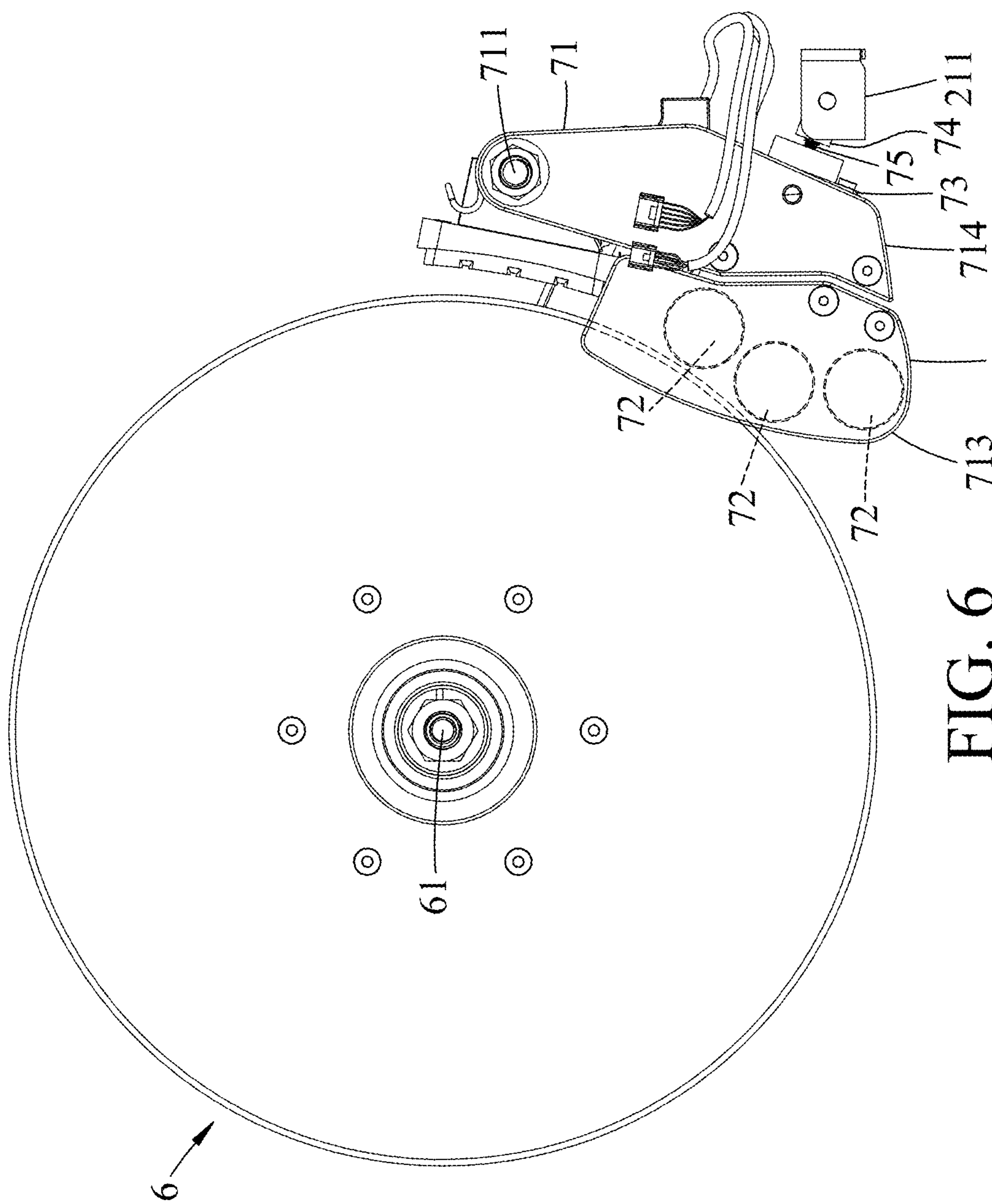


FIG. 6

STAIR-CLIMBING MACHINE AND RESISTANCE CONTROLLING DEVICE THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Taiwanese Invention patent application Ser. No. 11/310,8402, filed on Mar. 7, 2024, the entire disclosure of which is incorporated by reference herein.

FIELD

[0002] The disclosure relates to fitness equipment and a resistance device thereof, and more particularly to a stair-climbing machine and a resistance device thereof.

BACKGROUND

[0003] A conventional stair-climbing machine is a type of fitness equipment that allows a user to simulate stair climbing. The conventional stair-climbing machine is popular fitness equipment, which enhances cardiopulmonary functions, increases stamina, and strengthens leg muscles and gluteal muscles.

[0004] The conventional stair-climbing machine includes a plurality of stairs and a motor. When a user steps on the stairs, the motor provides a resistance force that opposes movement of the stairs. The magnitude of the resistance force can be changed by altering the electric current or rotation speed of the motor using a motor controlling system. However, the structures of the motor and the motor controlling system are complicated, large, heavy, and cost lots of money to build. Hence, there is room for improvement.

SUMMARY

[0005] Therefore, an object of the disclosure is to provide a resistance controlling device and a stair-climbing machine that can alleviate at least one of the drawbacks of the prior art.

[0006] According to one aspect of the disclosure, the resistance controlling device is adapted to be installed to a base frame unit of a stair-climbing machine. The resistance controlling device includes a flywheel and a damper unit. The flywheel is made of metal, and includes an axle extending in an axial direction along an axis and adapted to be mounted in the base frame unit of the stair-climbing machine. The flywheel is rotatable about the axis. The damper unit is adapted to be mounted in the base frame unit of the stair-climbing machine, and includes a damper rack and a plurality of permanent magnets disposed on the damper rack. The damper rack is movable between a high-resistance position, where the damper rack is proximate to the flywheel, and a low-resistance position, where the damper rack is distal from the flywheel. When the flywheel is driven to rotate as the damper rack is in the high-resistance position, the permanent magnets exert a first resistance force on the flywheel. When the flywheel is driven to rotate as the damper rack is in the low-resistance position, the permanent magnets exert a second resistance force on the flywheel. The second resistance force is weaker than the first resistance force.

[0007] Another aspect of the disclosure is to provide a stair-climbing machine including a base frame unit, a transmission device, a control device, and the abovementioned resistance controlling device.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

[0009] FIG. **1** is a perspective view illustrating an embodiment of a stair-climbing machine according to the disclosure.

[0010] FIG. **2** is a fragmentary side view of the embodiment.

[0011] FIG. **3** is a perspective view of a resistance controlling device of the embodiment.

[0012] FIGS. **4** to **6** are side views of the resistance controlling device, illustrating a damper rack of the embodiment moving from a high-resistance position to a low-resistance position.

DETAILED DESCRIPTION

[0013] It should be noted herein that for clarity of description, spatially relative terms such as "top," "bottom," "upper," "lower," "on," "above," "over," "downwardly," "upwardly" and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently (e.g., rotated 90 degrees or at other orientations) and the spatially relative terms used herein may be interpreted accordingly.

[0014] Referring to FIGS. **1** and **2**, an embodiment of a stair-climbing machine according to the disclosure is for simulating stair climbing, and includes a base frame unit **2**, a transmission device **3**, a control device **4**, and a resistance controlling device **5**.

[0015] The base frame unit **2** includes a bottom frame **21** that has a mounting portion **211**, a base frame **22** that is mounted to the bottom frame **21**, and two handrails **23** that are respectively disposed on a right side and a left side of the base frame **22**. The resistance controlling device **5** is mounted to the mounting portion **211**. The handrails **23** are adapted for a user to grip when they are using the stair-climbing machine to exercise.

[0016] The transmission device **3** includes a transmission unit **31** mounted to the base frame unit **2**, and a stair unit **32** disposed around the transmission unit **31**. The stair unit **32** is movable relative to the base frame unit **2**. The transmission unit **31** includes a plurality of transmission shafts **311** extending in an axial direction, a plurality of gears **312** respectively mounted to the transmission shafts **311**, and a plurality of connection belts **313** connected to the gears **312**. The connection belts **313** may be chains or belts. The stair unit **32** is operable for driving operation of the transmission unit **31** through the connection between the gears **312** and the connection belts **313**. Since the transmission device **3** is well known in the art, further description thereof will be omitted for the sake of brevity.

[0017] The control device **4** is mounted to the base frame unit **2**, and includes a control panel **41**. The resistance controlling device **5** is signally connected to the control device **4**. The control panel **41** is operable for outputting a control signal to control and adjust output of a resistance force from the resistance controlling device **5**.

[0018] The resistance controlling device **5** is installed to the base frame unit **2**. The resistance controlling device **5** includes a flywheel **6** and a damper unit **7**. The flywheel **6** is electrically conductive and made of metal, and includes an axle **61** extending in the axial direction along an axis and adapted to be mounted in the stair-climbing machine. The flywheel **6** is rotatable about the axis. The axle **61** is connected to one of the connection belts **313**, and the stair unit **32** is operable for driving the flywheel **6** to rotate via the transmission unit **31**. That is to say, when the stair unit **32** is driven to move, the stair unit **32** drives the transmission unit **31** to operate, which in turn drives the flywheel **6** to rotate.

[0019] Referring to FIGS. **3** to **6**, the damper unit **7** is mounted to the base frame unit **2** and is signally connected to the control device **4**. The damper unit **7** includes a damper rack **71**, a plurality of permanent magnets **72** that are disposed on the damper rack **71**, a motor **73** that is fixed to the damper rack **71**, a fixed member **74** that is mounted to the mounting portion **211**, and a screw rod **75** that interconnects the motor **73** and the fixed member **74**.

[0020] The damper rack **71** has a fixed end **712** and a pivot end **713** that are opposite to each other. The damper rack **71** includes a pivot shaft **711** that is disposed at the fixed end **712**, and that is mounted to the base frame unit **2**, so that the pivot end **713** is pivotable relative to the base frame unit **2** with the pivot shaft **711** serving as a pivot.

[0021] The damper rack **71** further includes a rack body **714** that extends from the fixed end **712** to the pivot end **713**, and two magnet mount plates **715** that are disposed at the pivot end **713** and that are spaced apart from each other in the axial direction. The flywheel **6** is disposed between the magnet mount plates **715**. The permanent magnets **72** are disposed on the magnet mount plates **715**. The pivot shaft **711** extends through the rack body **714** so that the rack body **714** is pivotable about the pivot shaft **711**.

[0022] Each of the magnet mount plates **715** has a surface facing the flywheel **6**, and the permanent magnets **72** are disposed on the surfaces of the magnet mount plates **715**. Each of the permanent magnets **72** is spaced apart from the flywheel **6** (i.e., the permanent magnets **72** are not in contact with the flywheel **6**). In this embodiment, the damper unit **7** has six permanent magnets **72** (only four are shown in FIG. **3**). Three of the permanent magnets **72** are disposed on the surface of one of the magnet mount plates **715**, and the remaining permanent magnets **72** are disposed on the surface of the other one of the magnet mount plates **715**. When the flywheel **6** is driven to rotate relative to the damper rack **71**, the relative movement between the permanent magnets **72** and the flywheel **6** results in electromagnetic induction and develops eddy currents on the flywheel **6**, causing the flywheel **6** to experience a resistance force that is generated by the permanent magnets **72** and that opposes the rotation of the flywheel **6**, which is similar to the principle of an eddy current brake (ECB), the principle of ECB stopping items from moving using electromagnetism. In other embodiments, the number of the permanent magnets **72** may be changed in accordance to the magnetic strength of the permanent magnets **72** and/or in order to meet a required resistance force.

[0023] The motor **73** is fixed to the pivot end **713**, and may be a step motor. The fixed member **74** is pivotably mounted to the mounting portion **211**. The motor **73** is operable for driving the screw rod **75** to spin and drive the motor **73** to move towards or away from the fixed member **74**.

[0024] Referring to FIGS. **4** to **6**, the damper rack **71** is movable between a high-resistance position (see FIG. **4**), where the pivot end **713** of the damper rack **71** is proximate to the flywheel **6**, and a low-resistance position (see FIG. **6**), where the pivot end **713** of the damper rack **71** is distal from the flywheel **6**. When the flywheel **6** is driven to rotate as the damper rack **71** is in the high-resistance position, the permanent magnets **72** exert a first resistance force on the flywheel **6**. When the flywheel **6** is driven to rotate as the damper rack **71** is in the low-resistance position, the permanent magnets **72** exert a second resistance force on the flywheel **6**. A portion of the flywheel **6** that overlaps the permanent magnets **72** in the axial direction when the damper rack **71** is in the high-resistance position is larger than a portion of the flywheel **6** that overlaps the permanent magnets **72** in the axial direction when the damper rack **71** is in the low-resistance position, therefore the second resistance force being weaker than the first resistance force. When the damper rack **71** is moved from the high-resistance position to the low-resistance position, the resistance force from the permanent magnets **72** is reduced. When the damper rack **71** is moved from the low-resistance position to the high-resistance position, the resistance force from the permanent magnets **72** is increased. It should be noted that, when the damper rack **71** is in the low-resistance position, the flywheel **6** may not overlap the permanent magnets **72** in the axial direction. In this embodiment, no portions of the flywheel **6** overlap the permanent magnets **72** in the axial direction when the damper rack **71** is in the low-resistance position. Furthermore, by virtue of the configurations of the motor **73**, the fixed member **74** and the screw rod **75**, when the motor **73** is driven to move towards the fixed member **74**, the damper rack **71** is driven to move to the low-resistance position; and when the motor **73** is driven to move away from the fixed member **74**, the damper rack **71** is driven to move to the high-resistance position.

[0025] Referring to FIGS. **1** and **4**, a process of using the stair-climbing machine is as follows. A user operates the control panel **41** and outputs a control signal to adjust output of the resistance force from the resistance controlling device **5**. Then, the damper unit **7** receives the control signal, and moves the damper rack **71** of the damper unit **7** between the high-resistance position and the low-resistance position in accordance to the control signal to respectively output the first resistance force and the second resistance force.

[0026] Specifically, the control panel **41** is operable for outputting a control signal for high resistance output and another control signal for low resistance output. When the control panel **41** is operated to output the control signal for high resistance output, the damper unit **7** receives the control signal, and drives the motor **73** to drive the screw rod **75** to spin and drive the motor **73** to move away from the fixed member **74**, which drives the damper rack **71** to move to the high-resistance position, and increases portions of the flywheel **6** that overlap the permanent magnets **72** in the axial direction. At this time, when the flywheel **6** is driven to rotate, the permanent magnets **72** exert the first resistance force onto the flywheel **6** to oppose rotation of the flywheel **6**. By virtue of the configurations of the flywheel **6** and the stair unit **32**, a resistance force opposing movement of the stair unit **32** is increased.

[0027] When the control panel **41** is operated to output the another control signal for low resistance output, the damper unit **7** receives the another control signal, and drives the motor **73** to drive the screw rod **75** to spin and drive the motor **73** to move towards the fixed member **74**, which drives the damper rack **71** to move to the low-resistance position, and which decreases portions of the flywheel **6** that overlap the permanent magnets **72** in the axial direction. At this time, when the flywheel **6** is driven to rotate, the permanent magnets **72** exert the second resistance force onto the flywheel **6** to oppose rotation of the flywheel **6**.

[0028] By virtue of the configurations of the flywheel **6** and the stair unit **32**, a resistance force opposing movement of the stair unit **32** is decreased. In conclusion, advantages of the disclosure are as follows.

[0029] The flywheel **6** is made of metal, and the permanent magnets **72** are disposed on the damper rack **71**; since the damper rack **71** is movable relative to the flywheel **6**, when the flywheel **6** is driven to rotate, by moving the damper rack **71** to the high-resistance position or the low-resistance position, the permanent magnets **72** can respectively exert the first resistance force or the second resistance force on the flywheel **6**. In comparison to the aforementioned conventional stair-climbing machine, the structure of the stair-climbing machine is simpler, and the size, the weight and the manufacturing costs of the stair-climbing machine are reduced, while the resistance output of the stair-climbing machine can still be adjusted.

[0030] Through the small structure of the damper rack **71**, by pivoting the rack body **714** relative to the pivot shaft **711**, portions of the flywheel **6** that overlap the permanent magnets **72** in the axial direction can be adjusted. Hence, the size and the weight of the stair-climbing machine are reduced compared to the conventional stair-climbing machine.

[0031] Through the simple structure of the motor **73**, the screw rod **75** and the fixed member **74**, the damper rack **71** can be moved between the high-resistance position and the low-resistance position. Furthermore, the simple structure is easy to maintain, which reduces costs.

[0032] In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to "one embodiment," "an embodiment," an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

[0033] While the disclosure has been described in connection with what is(are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

What is claimed is:

1. A resistance controlling device being adapted to be installed to a base frame unit of a stair-climbing machine, said resistance controlling device comprising:

a flywheel that is made of metal, that includes an axle extending in an axial direction along an axis and adapted to be mounted in the base frame unit of the stair-climbing machine, and that is rotatable about the axis; and

a damper unit that is adapted to be mounted in the base frame unit of the stair-climbing machine, and that includes a damper rack and a plurality of permanent magnets disposed on said damper rack;

wherein, said damper rack is movable between a high-resistance position, where said damper rack is proximate to said flywheel, and a low-resistance position, where said damper rack is distal from said flywheel;

wherein, when said flywheel is driven to rotate as said damper rack is in the high-resistance position, said plurality of permanent magnets exert a first resistance force on said flywheel; and

wherein, when said flywheel is driven to rotate as said damper rack is in the low-resistance position, said plurality of permanent magnets exert a second resistance force on said flywheel, the second resistance force being weaker than the first resistance force.

2. The resistance controlling device as claimed in claim **1**, wherein:

said damper rack has a fixed end and a pivot end that are opposite to each other;

said damper rack includes a pivot shaft that is disposed at said fixed end, and that is adapted to be mounted to the base frame unit of the stair-climbing machine, so that said pivot end is adapted to be pivotable relative to the base frame unit of the stair-climbing machine, with said pivot shaft serving as a pivot, to convert said damper rack between the high-resistance position and the low-resistance position;

when said damper rack is in the high-resistance position, said pivot end is proximate to said flywheel;

when said damper rack is in the low-resistance position, said pivot end is distal from said flywheel; and

a portion of said flywheel that overlaps said plurality of permanent magnets in the axial direction when said damper rack is in the high-resistance position is larger than a portion of said flywheel that overlaps said plurality of permanent magnets in the axial direction when said damper rack is in the low-resistance position.

3. The resistance controlling device as claimed in claim **2**, wherein:

said damper rack further includes a rack body that extends from said fixed end to said pivot end, said pivot shaft extending through said rack body so that said rack body is pivotable about said pivot shaft; and

said damper rack further includes two magnet mount plates that are disposed at said pivot end and that are spaced apart from each other in the axial direction, said flywheel being disposed between said magnet mount plates, said plurality of permanent magnets being disposed on said magnet mount plates.

4. The resistance controlling device as claimed in claim **3**, wherein each of said magnet mount plates has a surface facing said flywheel, said plurality of permanent magnets being disposed on said surfaces of said magnet mount plates.

5. The resistance controlling device as claimed in claim **2**, wherein:

said damper unit further includes a motor that is fixed to said pivot end of said damper rack, a fixed member that is adapted to be mounted to the base frame unit of the stair-climbing machine, and a screw rod that interconnects said motor and said fixed member;

said motor is operable for driving said screw rod to spin and drive said motor to move towards or away from said fixed member;

when said motor is driven to move towards said fixed member, said damper rack is driven to move to the low-resistance position; and

when said motor is driven to move away from said fixed member, said damper rack is driven to move to the high-resistance position.

6. A stair-climbing machine comprising:

a base frame unit;

a transmission device that includes

a transmission unit mounted to said base frame unit, and

a stair unit disposed around said transmission unit, being movable relative to said base frame unit, and being operable for driving operation of said transmission unit;

a control device that is mounted to said base frame unit and that is operable for outputting a control signal; and

said resistance controlling device as claimed in claim **1**, said flywheel of said resistance controlling device being connected to said transmission unit, said damper unit of said resistance controlling device being mounted to said base frame unit and being signally connected to said control device;

wherein said damper unit receives the control signal, and moves said damper rack of said damper unit between the high-resistance position and the low-resistance position in accordance to the control signal; and

wherein, said stair unit is operable for driving, via said transmission unit, said flywheel to rotate.

7. The stair-climbing machine as claimed in claim **6**, wherein:

said damper rack has

a fixed end and a pivot end that are opposite to each other;

said damper rack includes a pivot shaft that is disposed at said fixed end, and that is mounted to said base frame unit, so that said pivot end is pivotable relative to said base frame unit, with said pivot shaft serving as a pivot, to convert said damper rack between the high-resistance position and the low-resistance position;

when said damper rack is in the high-resistance position, said pivot end is proximate to said flywheel;

when said damper rack is in the low-resistance position, said pivot end is distal from said flywheel; and

a portion of said flywheel that overlaps said plurality of permanent magnets in the axial direction when said damper rack is in the high-resistance position is larger than a portion of said flywheel that overlaps said plurality of permanent magnets in the axial direction when said damper rack is in the low-resistance position.

8. The stair-climbing machine as claimed in claim **7**, wherein:

said damper rack further includes a rack body that extends from said fixed end to said pivot end, said pivot shaft extending through said rack body so that said rack body is pivotable about said pivot shaft; and

said damper rack further includes two magnet mount plates that are disposed at said pivot end and that are spaced apart from each other in the axial direction, said flywheel being disposed between said magnet mount plates, said plurality of permanent magnets being disposed on said magnet mount plates.

9. The stair-climbing machine as claimed in claim **8**, wherein each of said magnet mount plates has a surface facing said flywheel, said plurality of permanent magnets being disposed on said surfaces of said magnet mount plates.

10. The stair-climbing machine as claimed in claim **7**, wherein

said damper unit further includes a motor that is fixed to said pivot end of said damper rack, a fixed member that is mounted to said base frame unit, and a screw rod that interconnects said motor and said fixed member;

said motor is operable for driving said screw rod to spin and drive said motor to move towards or away from said fixed member;

when said motor is driven to move towards said fixed member, said damper rack is driven to move to the low-resistance position; and

when said motor is driven to move away from said fixed member, said damper rack is driven to move to the high-resistance position.

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